

Multiplicity: Operator Dictionary (State = (x, H))

This document specifies (a) a **precise operator dictionary** where each symbol denotes a total/partial function on a state (x, H) , and (b) a **concrete experimental protocol** (datasets, baselines, ablations) aligned with the blueprint's predictions.

0) Notation and state space

Cyclic index set

- Fix $n \in \mathbb{N}$ with prime factorization $n = \prod_i p_i^{a_i}$.
- Let $G := \mathbb{Z}/n\mathbb{Z}$ (indices t are taken mod n).

Signal component

- Let d_x be the signal dimension.
- A signal is $x : G \rightarrow \mathbb{R}^{d_x}$. Equivalently, $x \in \mathbb{R}^{n \times d_x}$ with rows indexed by $t \in G$.

Hypergraph component

Use an attributed, role-aware hypergraph:

- $H := (V, E, \iota, \rho, a_V, a_E)$
- V : finite vertex set.
- E : finite hyperedge set.
- $\iota : E \rightarrow \text{List}(V)$: incidence **list** (ordered) of vertices per edge.
- $\rho : E \rightarrow \text{List}(\mathcal{R})$: role labels aligned with $\iota(e)$ (same length).
- $a_V : V \rightarrow \mathbb{R}^{d_V}$: vertex attributes.
- $a_E : E \rightarrow \mathbb{R}^{d_E}$: edge attributes.

Using ordered incidence (lists) + roles makes n-ary relations explicit.

Full state

- **State** $:= X \times \mathcal{H}$ where $X = L^2(G; \mathbb{R}^{d_x})$ (finite-dimensional here) and $\mathcal{H}$ is the space of such hypergraphs.
- Operators are functions **State** $\rightarrow$ **State**, possibly **partial** (defined only when preconditions hold).

Convention

For an operator $\mathcal{O}$:

- $\mathcal{O}(x, H) = (x', H')$.
- If $\mathcal{O}$ acts only on one component, the other is unchanged.

1) Prime-power “level” operators on signals

We define level operators using subgroup-averaging on G .

1.1 Subgroups and coset averaging

For each divisor $d \mid n$ define subgroup

$$H_d := \{0, d, 2d, \dots, (\frac{n}{d}-1)d\} \subseteq G,$$

$$\text{so } |H_d| = n/d.$$

Define the averaging kernel

$$\mu_{H_d} := \frac{1}{|H_d|} \sum_{h \in H_d} \delta_h.$$

Symbol: Π_d (in the blueprint: Π_{p^r} for prime powers)

Definition on signals:

$$(\Pi_d x)(t) := (x * \mu_{H_d})(t) = \frac{1}{|H_d|} \sum_{h \in H_d} x(t-h).$$

Action on state:

$$\Pi_d(x, H) := (\Pi_d x, H).$$

Properties (useful for pruning/normal forms):

- Idempotent: $\Pi_d \circ \Pi_d = \Pi_d$.
- Contractive in ℓ_2 .

Prime-power specialization: Use $d = p^r$ whenever $p^r \mid n$.

Symbol: Δ_d (detail), Sp_d (spike)

Two common choices (pick one and keep it consistent):

(A) Residual-to-level:

$$\Delta_d := I - \Pi_d, \quad \text{Sp}_d := I - \Pi_d.$$

(B) Band-pass between nested levels when $d_1 \mid d_2$:

$$\Delta_{d_1 \text{ to } d_2} := \Pi_{d_2} - \Pi_{d_1}.$$

Action on state:

$$\Delta_d(x, H) := (\Delta_d x, H), \quad \mathrm{Sp}_d(x, H) := ((I - \Pi_d)x, H). \quad \square$$

1.4 Level selection by CRT (optional but aligned with blueprint)

Let $n = \prod_i p_i^{a_i}$. You may define a **multi-indexed projector**

$$\Pi_{\mathbf{r}} := \Pi_{p_1^{r_1}} \circ \dots \circ \Pi_{p_k^{r_k}}, \quad 0 \leq r_i \leq a_i. \quad \square$$

Even if the Π 's commute, this still provides a clean lattice of levels for search.

2) Group actions and “accent” operators on signals

Symbol: R_k

Definition:

$$(R_k x)(t) := x(t+k). \quad \square$$

Action on state:

$$R_k(x, H) := (R_k x, H). \quad \square$$

If your hypergraph has time-indexed vertices (see §4.3), define a synchronized version $\tilde{R}_k$ that also shifts those vertex IDs.

Symbol: A (parameterized)

Let $\chi : G \rightarrow \mathbb{R}$ (or S^1 if you keep complex scalars). Typical choices:

- Fourier character: $\chi_m(t) = \cos(2\pi mt/n)$ and/or $\sin(2\pi mt/n)$.
- Learned gate: $\chi(t) = \sigma(u^\top x(t) + b)$ (keep this fixed during operator definition; learning is outside the operator).

Definition:

$$(A_\chi x)(t) := \chi(t) x(t) \quad \text{(scalar gate)} \quad \square$$

or coordinate-wise gating $(A_\chi x)(t)_j = \chi_j(t) x(t)_j$.

Action on state: $A_\chi(x, H) = (A_\chi x, H)$.

Symbol: W (parameterized)

Let $\varphi : G \rightarrow G$ be a bijection. A canonical family is multiplication by q with $\gcd(q, n) = 1$: $\varphi_q(t) = qt \bmod n$.

Definition:

$$W_{\varphi}(x)(t) := x(\varphi^{-1}(t)).$$

Action on state: $W_{\varphi}(x, H) = (W_{\varphi}x, H)$ unless coupled to time vertices (see §4.3).

W is the main source of noncommutativity with Π_d in practice: W changes the partition/cosets being averaged.

3) Hypergraph operators (typed rewrites)

All hypergraph operators are defined as functions on H with explicit preconditions. Extend to state by leaving x unchanged unless stated.

Symbol: W (hypergraph-side)

Let $\sigma : V \rightarrow V$ be a bijection.

Definition:

- $V' = V, E' = E$
- $\iota'(e) = \text{map}(\sigma, \iota(e))$
- $a'_V(\sigma(v)) = a_V(v)$
- Edge roles/attrs unchanged.

State action: $W_{\sigma}(x, H) = (x, W_{\sigma}(H))$.

Symbol: $S_{v,P}$

Parameters:

- vertex $v \in V$
- predicate P on incident edge-positions: $P : (e, i) \mapsto \{0, 1\}$ for $\iota(e)[i] = v$

Precondition: $v \in V$.

Definition: create two new vertices v_0, v_1 with attributes derived from $a_V(v)$:

- $V' = (V \setminus \{v\}) \cup \{v_0, v_1\}$
- For each incident position (e, i) where $\iota(e)[i] = v$, set
- $\iota'(e)[i] = v_{P(e,i)}$

- Set $a'_V(v_b) = \text{SplitAttr}(a_V(v), b)$ (e.g., copy or learned affine maps).

State action: $S_{v,P}(x, H) = (x, S_{v,P}(H))$.

Symbol: $M_{u,v}$

Precondition: $u \neq v \in V$.

Definition: merge u and v into w :

- $V' = (V \setminus \{u, v\}) \cup \{w\}$
- Replace occurrences of u or v in ι by w
- $a'_V(w) = \text{MergeAttr}(a_V(u), a_V(v))$ (e.g., mean/concat+MLP).

State action: $M_{u,v}(x, H) = (x, M_{u,v}(H))$.

3.4 Edge split / merge (optional)

Often useful for n-ary relations.

Edge split $S_{e,Q}^E$: split an edge e into e_0, e_1 based on predicate $Q(i)$ over positions.

Edge merge M_{e_1,e_2}^E : merge if they share schema/roles.

Symbol: F

Transforms hypergraph into bipartite incidence form (useful as a normal form / baseline):

- Create a vertex v_e for each edge e
- Replace each hyperedge by ordinary edges $(v_e, \iota(e)[i])$ carrying role $\rho(e)[i]$

Denote $F(H)$ as this factor graph. This is a **deterministic compilation**.

State action: $F(x, H) = (x, F(H))$.

Symbol: Rel_τ

Let τ be a schema transform mapping roles/edge-types.

Definition: re-map ρ and optionally partition edges by type.

Symbol: U_{f_V, f_E}

Functions:

- $f_V : \mathbb{R}^{d_V} \rightarrow \mathbb{R}^{d_V}$

- $f_E : \mathbb{R}^{d_E} \rightarrow \mathbb{R}^{d_E}$

Definition: $a'_V(v) = f_V(a_V(v))$, $a'_E(e) = f_E(a_E(e))$.

3.8 Add/remove edges (structural edits)

Add: $E^+_{(v_1, \dots, v_k), (r_1, \dots, r_k), \alpha}$

- Adds new edge with incidence list and roles, edge attribute α .

Remove: E^-_e

- Removes edge e .

These are powerful; in many experiments, restrict to **typed templates** to avoid combinatorial blowup.

4) Coupling operators between signal and hypergraph

These are the glue that makes (x, H) a single system rather than two disjoint objects.

Symbol: RO_g

Produces a task output y (classification/regression/ranking) from state:

$$\text{RO}_g(x, H) := g(\text{Pool}_G(x), \text{Pool}_H(H))$$

where pooling can be mean/max/attention, and g is a simple head (linear/MLP).

Readout is not necessarily included in the operator word (you can treat it as the final evaluation stage).

Symbol: Lift_ψ

Given a map $\psi : V \rightarrow G$ (assign each vertex a time index), define a lifted signal

$$(\text{Lift}_\psi(H))(t) := \text{Agg}_{a_V(v): \psi(v)=t} \mathbb{R}^{d_V}.$$

Then combine with x via concatenation:

$$\widetilde{x}(t) := \text{concat}(x(t), \text{Lift}_\psi(H)(t)).$$

State action: $\text{Lift}_\psi(x, H) = (\widetilde{x}, H)$.

If $\psi : V \rightarrow G$ is part of your state, define:

- $\tilde{R}_k$ shifts x by k and reindexes time-assigned vertices so $\psi'(v) = \psi(v) + k$.

This is often necessary to make R_k meaningful when H encodes events over time.

5) Operator words, typing, and normal forms

5.1 Words

An **operator word** is a composition

$$W := \mathcal{O}_1 \circ \dots \circ \mathcal{O}_2 \circ \mathcal{O}_1$$

where each $\mathcal{O}_i$ is a parameterized operator from the dictionary.

5.2 Admissibility (lightweight refinement typing)

Each operator includes:

- **Precondition** $\text{Pre}_{\mathcal{O}}(x, H)$
- **Postcondition/invariants** $\text{Inv}(x, H)$

A word is **admissible** if all preconditions hold along the execution trace and invariants are preserved.

5.3 Normal forms (for search pruning)

Use algebraic simplifications to reduce branching:

- $\Pi_d \Pi_d \rightarrow \Pi_d$
 - $(I - \Pi_d) \Pi_d \rightarrow 0$ (if you represent it explicitly)
 - Optional: collapse consecutive shifts $R_a R_b \rightarrow R_{a+b}$
 - Optional: if you restrict Π to a commuting family, reorder them into canonical order.
-

(b) Experimental Protocol

6) Core experimental question

Does a **prime-level operator word** (noncommutative composition) produce measurable gains in:

1. periodic anomaly detection,
2. n-ary / hypergraph structure learning,
3. retrieval interference reduction,

4. interpretability stability, compared to commutative or unconstrained baselines?

7) Implementation skeleton

Pick n to have multiple primes (e.g., $n = 2^a 3^b 5^c$). Practical defaults:

- $n = 720 (2^4 \cdot 3^2 \cdot 5)$ for minute/hour/day structure
- $n = 840 (2^3 \cdot 3 \cdot 5 \cdot 7)$ for richer prime mix

For a time series $z_1, \dots, z_T$, create overlapping windows of length n :

- Window w corresponds to $x_w(t) = \text{feat}(z_{w+t}) \in \mathbb{R}^{d_x}$

Hypergraph H_w can be:

- trivial (empty) for pure time-series tasks,
- event-derived (edges = events within window),
- knowledge-derived (edges = facts relevant to window).

7.2 Word search (beam)

- Alphabet Σ : start small
- signal: $\Pi_{p^r}, \Delta_{p^r}, R_{\pm 1}, A_{\chi_m}, W_{\varphi_q}$
- hypergraph: S, M, F, U (optional in phase 1)
- Max length $L \in \{6, 8, 10\}$
- Beam width $B \in \{64, 128, 256\}$
- Score = validation resonance (metric-specific) minus complexity penalty $\lambda \cdot L$

Pruning:

- Apply normal forms (§5.3)
- Reject inadmissible words early (failed preconditions)

7.3 Resonance metrics by task

- Anomaly detection: AUPRC / AUROC / F1 at fixed false-positive rate
 - Hyper-relational prediction: MRR, Hits@K
 - Retrieval interference: accuracy vs load + latency; interference slope
 - Interpretability: stability metrics (edit distance, Jaccard over operator multiset, agreement of top-k attributions)
-

8) Experiment A — Periodic anomaly detection

8.1 Datasets

Use a mix of synthetic + real-world benchmarks:

- NAB (streaming anomaly detection)
- Yahoo S5 (A1–A4)
- AIOps-style KPI datasets (multivariate seasonality)
- Industrial control (SWaT / WADI) if available

8.2 Task

Given a window x_w , predict whether it contains an anomaly (binary) or produce pointwise anomaly scores.

8.3 Multiplicity model (minimal)

- Build features by applying word W to x : $\phi_W(x) = \text{Pool}_G(W(x))$
- Classifier: logistic regression / small MLP
- Search W with beam search; choose best on validation.

8.4 Baselines

Feature baselines:

- FFT magnitude features + linear model
- Wavelet features + linear model
- Seasonal decomposition + residual threshold

Learning baselines:

- Isolation Forest / One-Class SVM on standard features
- LSTM/TCN autoencoder anomaly score
- Transformer-based anomaly detection (where feasible)

8.5 Ablations (directly aligned to predictions)

1. **No prime levels:** remove Π_p ; allow only shifts/accents.
2. **Commutative aggregation:** compute features $\sum_i f_i(x)$ from operator set, ignore order.
3. **Order-only:** keep same multiset of operators but randomize order; compare.
4. **No permutation ***** W *****:** disallow index automorphisms.
5. **No accents ***** A *****:** disallow modulation gates.
6. **Fixed hand-crafted word:** e.g., Π_p and Δ_p only.

8.6 Expected measurable outcome

- Noncommutative word search should improve AUPRC/AUROC over (2) commutative aggregation, especially when multiple periodicities exist.
-

9) Experiment B — Hypergraph structure learning (n-ary / hyper-relational)

9.1 Datasets

Choose datasets with explicit n-ary relations / qualifiers:

- Hyper-relational knowledge graphs (statements with qualifiers)
- N-ary relation corpora (entities + roles)

Concrete options commonly used in the literature:

- WikiPeople (n-ary facts)
- JF17K (n-ary)
- WD50K variants (Wikidata statements)

9.2 Task variants

(i) **Missing argument prediction:** given edge e with one role missing, predict the missing entity.

(ii) **Edge validation:** classify whether a proposed n-ary fact is valid.

9.3 Multiplicity model (hypergraph-enabled)

State includes H for the window/document:

- H nodes = entities; hyperedges = facts; roles = argument positions.
- Optionally define x as an auxiliary signal (e.g., time ordering of facts or textual position indices).

Apply a word W consisting of:

- structural rewrites S, M (restricted templates)
- compilation F (optional)
- attribute updates U

Readout for prediction:

- Score candidates by RO over transformed H' (and x' if used).

9.4 Baselines

- Factor-graph/bipartite compilation (use F) + standard GNN message passing
- Hypergraph neural networks (HGNN-style / HyperGCN-style) on the incidence structure
- Hyper-relational KG baselines (statement/qualifier models)

9.5 Ablations

1. **No rewrites:** only U on attributes; structure fixed.
2. **No role awareness:** drop ρ (treat incidence as unordered set).
3. **Only compilation** ***** F *****: reduce to factor graph; no higher-order ops.
4. **No prime-level coupling:** disable Lift/time alignment if used.
5. **Rewrite-only vs attribute-only:** isolate where gains come from.

9.6 Metrics

- MRR, Hits@1/3/10 for argument prediction
- AUROC/AUPRC for edge validation
- Constraint violation rate (typed schema errors) if you enforce invariants

9.7 Expected measurable outcome

- Typed rewrites should reduce structural distortion/constraint violations vs unconstrained edits, while maintaining or improving MRR/Hits@K.

10) Experiment C — Retrieval interference in memory/KG

10.1 Setup (controlled)

Create a memory of items $\{m_i\}$ as hyperedges and/or signal patterns.

- Queries are partial patterns; retrieval returns top-k matches.

Increase load N and measure:

- accuracy@k vs N
- latency vs N
- interference slope (performance drop per doubling of N)

10.2 Multiplicity approach

Use prime-level projections as **orthogonalizing/partitioning** features:

- Store each item with a signature across Π_{p^r} levels.
- Query uses the same operator word for feature extraction.

10.3 Baselines

- Dense embedding retrieval (vector search)
- Hashing / LSH
- Graph traversal / rule-based retrieval

10.4 Ablations

- Remove prime-level decomposition
- Replace Π_{p^r} with single-scale smoothing

10.5 Expected outcome

- Prime-level signatures reduce cross-talk, improving accuracy under heavy load and/or improving latency via early pruning.

11) Experiment D — Interpretability and stability

11.1 What is “the explanation”?

Multiplicity’s explanation is the selected word W^* plus its intermediate states (x_i, H_i) .

11.2 Stability protocol

Train/select W^* across S random seeds or folds. Compare:

- edit distance between words
- Jaccard similarity of operator multisets
- agreement of top contributing operators (by ablation impact)

11.3 Baseline explanations

- Feature attribution on FFT/wavelet features
- Gradient/attention-based explanations (for neural baselines)

11.4 Expected outcome

- Operator-level explanations should be more stable across runs than dense attribution maps, especially when you constrain words via admissibility.

12) Minimal “Phase 1” plan (fastest validation)

1. Implement signal operators: $\Pi_{p^r}, \Delta_{p^r}, R_{\pm 1}, A_{\chi_m}, W_{\varphi_q}$.
2. Run Experiment A (anomaly detection) with beam search over words up to length 8.
3. Report: best word, ablation table, runtime, and whether ordering beats commutative aggregation.

4. Only if successful: add hypergraph operators S, M, F, U and run Experiment B.

Appendix: Operator checklist (symbols $\rightarrow$ function)

Signal-side:

- Π_{p^r} : coset averaging projector
- Δ_{p^r} : residual or band-pass
- R_k : shift
- A_χ : modulation/gating
- W_φ : index permutation/automorphism

Hypergraph-side:

- $S_{v,P}$: vertex split
- $M_{u,v}$: vertex merge
- F : edge-to-vertex fold (factorization)
- Rel_τ : role/schema rewrite
- U_{f_V, f_E} : attribute update
- E^+, E^- : add/remove edges

Coupling:

- Lift_ψ : hypergraph-to-signal lift
- $\hat{R}_k$: synchronized shift (signal + time-indexed vertices)
- RO_g : readout (task head)

References

1. *Ryan Van Gelder “*Multiplicity: Type-theoretic Kr Blueprint*” (unpublished manuscript / blueprint PDF).
2. Alexander Lavin and Subutai Ahmad. “Evaluating Real-time Anomaly Detection Algorithms — The Numenta Anomaly Benchmark.” arXiv:1510.03336 (2015).
3. Yahoo! Webscope. “S5 — A Labeled Anomaly Detection Dataset, Version 1.0 (16M).” (2015).
4. Aditya P. Mathur and Nils Ole Tippenhauer. “SWaT: A Water Treatment Testbed for Research and Training on ICS Security.” Proceedings of the Workshop on Cyber-Physical Systems for Smart Water Networks (CySWater), April 2016.
5. Chuadhry Mujeeb Ahmed, Venkata Reddy Palleti, and Aditya P. Mathur. “WADI: A Water Distribution Testbed for Research in the Design of Secure Cyber Physical Systems.” CySWater (2017).
6. Jiehui Xu, Haixu Wu, Jianmin Wang, and Mingsheng Long. “Anomaly Transformer: Time Series Anomaly Detection with Association Discrepancy.” arXiv:2110.02642 (2021).
7. Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou. “Isolation Forest.” 2008 Eighth IEEE International Conference on Data Mining (ICDM), 413–422 (2008).
8. Bernhard Schölkopf, John C. Platt, John Shawe-Taylor, Alex J. Smola, and Robert C. Williamson. “Estimating the Support of a High-Dimensional Distribution.” *Neural Computation* 13(7):1443–1471 (2001).

9. Sepp Hochreiter and Jürgen Schmidhuber. "Long Short-Term Memory." *Neural Computation* 9(8):1735–1780 (1997).
10. Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. "Attention Is All You Need." *NeurIPS* (2017).
11. Jianfeng Wen, Jianxin Li, Yongyi Mao, Shini Chen, and Richong Zhang. "On the Representation and Embedding of Knowledge Bases Beyond Binary Relations." *IJCAI* 2016, 1300–1307 (2016).
12. Saiping Guan, Xiaolong Jin, Yuanzhuo Wang, and Xueqi Cheng. "Link Prediction on N-ary Relational Data." *Proceedings of the 28th International Conference on World Wide Web (WWW '19)*, 583–593 (2019).
13. Saiping Guan, Xiaolong Jin, Jiafeng Guo, Yuanzhuo Wang, and Xueqi Cheng. "NeuInfer: Knowledge Inference on N-ary Facts." *ACL* 2020, 6141–6151 (2020).
14. Mikhail Galkin, Priyansh Trivedi, Gaurav Maheshwari, Ricardo Usbeck, and Jens Lehmann. "Message Passing for Hyper-Relational Knowledge Graphs." *EMNLP* 2020, 7346–7359 (2020).
15. Bahare Fatemi, Perouz Taslakian, David Vazquez, and David Poole. "Knowledge Hypergraphs: Prediction Beyond Binary Relations." *IJCAI* 2020.
16. Naganand Yadati, Madhav Nimishakavi, Prateek Yadav, Vikram Nitin, Anand Louis, and Partha Talukdar. "HyperGCN: A New Method of Training Graph Convolutional Networks on Hypergraphs." *NeurIPS* 2018 / arXiv:1809.02589.
17. Yifan Feng, Haoxuan You, Zizhao Zhang, Rongrong Ji, and Yue Gao. "Hypergraph Neural Networks." *AAAI* 2019.
18. Ingrid Daubechies. *Ten Lectures on Wavelets*. SIAM (1992).
19. Stéphane G. Mallat. "A Theory for Multiresolution Signal Decomposition: The Wavelet Representation." *IEEE TPAMI* 11(7):674–693 (1989).
20. Patrick M. Rondon, Ming Kawaguchi, and Ranjit Jhala. "Liquid Types." *PLDI* 2008.
21. Nikhil Swamy, Juan Chen, Cédric Fournet, Pierre-Yves Strub, Karthikeyan Bhargavan, and Jean Yang. "Dependent Types and Multi-monadic Effects in F*." *POPL* 2016.
22. Hartmut Ehrig, Karsten Ehrig, Ulrike Prange, and Gabriele Taentzer. *Fundamentals of Algebraic Graph Transformation*. Springer (2006).
23. Benjamin C. Pierce. *Types and Programming Languages*. MIT Press (2002).